RPMGLOBAL

AMT Solution Design & Deployment Approach

August 2019



Document Control Sheet

Client	
Glencore Lomas Bayas and Glencore Antapaccay	
Document Name	**Date**
AMT Solution Design and Deployment Approach	28-Aug-19
File Name:	
Glencore - AMT Solution Design and Deployment Approach 20190828.pdf	

	Name	Position	Date
Prepared By:	RPMGlobal		28-Aug-19
Reviewed By:	Glencore		

Version	Date	Author	Comments
1.0	28-Aug-19	RPMGlobal	Initial version

Table of Contents

1. Overview 5
   1.1. Background 5
   1.2. Why a single instance? 6
      RPM Single Instance 6
      1.2.1. Feedback 6
   1.3. Why Go-live at 1st January 2020? 6
      1.3.1. RPM Go-live Feedback 6
2. AMT Solution Design 8
   2.1. Work Management Processes 8
      2.1.1. Asset Registration 8
      2.1.2. Strategy Definition 9
      2.1.3. Long-term Planning 9
      2.1.4. Asset Health 9
      2.1.5. Inspections 9
      2.1.6. Short-term Planning 10
      2.1.7. Downtime 10
      2.1.8. Work Execution 11
      2.1.9. Work Finalization 11
      2.1.10. Asset Metrics 11
      2.1.11. Warranty Identification 11
   2.2. Decision Support Processes 12
      2.2.1. Strategy Optimization 12
      2.2.2. Performance Analytics 12
      2.2.3. Product Problem Management 12
      2.2.4. Modelling 12
      2.2.5. Benchmarking 12
      2.2.6. Life-cycle Management 12
      2.2.7. Maintenance Budgeting 12
   2.3. Management Reporting 13
      2.3.1. Life-cycle Cost Reporting 13
      2.3.2. Performance Reporting 13
   2.4. System Integration 14
      2.4.1. Integration Data Flows and Descriptions 14
      2.4.2. Standardisation 16
   2.5. AMT System Data Entities 16
   2.6. AMT Software Enhancements 17

3. Solution Configuration 18
3.1. AMT Functionality 18
3.2. AMT User Security-level Configuration 19
3.2.1. Groups 19
3.2.2. User Accounts 19
3.2.3. Global Data Visibility and Access 19
3.3. AMT System-level Configuration 19
3.4. AMT System Table-level Configuration 21
3.5. IT Architecture 23
3.5.1. Time Zone 24
3.5.2. Time Zone Impact 24
3.5.3. Impact of server setting for daylight saving decision 25
4. Solution Deployment 26
4.1. Operations, Departments, Assets 26
4.2. IT Architecture 26
4.3. Data Preparation 26
4.4. User Training 27
4.5. Standardisation 27
4.6. Change Management 27
4.7. Rollout Approach 27
4.8. Glencore Project Team 28
5. Project Risks 29
6. Approval Section 31

List of Tables

Table 1 AMT Reporting Feedback 13
Table 2 Interface Descriptions 15
Table 3 System Data Entities 16
Table 4 Functionality to be Deployed 18
Table 5 AMT Configuration 20
Table 6 System Data Entities 22
Table 8 Project Risks 29
Table 9 AMT Solution Design Approval 31

List of Figures

Figure 1 AMT Solution and Functions 8
Figure 2 AMT Integration Solution Design Framework 14

1. Overview

Glencore and RPMGlobal ("RPM") held a workshop in Santiago on 5th and 6th of August 2019. The purpose of the workshop was to investigate an option to implement a single AMT Production instance/platform that would be accessed by both Antapaccay ("AP") and Lomas Bays ("LB").

The outcomes of the workshop are documented in section 2, 3, 4 and 5. The implementation proposal, for the solution, will be submitted by RPM in a separate document. It is based on the contents of this document.

1.1. Background

In September 2018, a scoping workshop was held in Peru, attended by personnel from RPM, LB and AP. The purpose of the workshop was to define how AP and LB would implement AMT. The main outcomes of the workshop were:

- LB and AP agreed to deploy AMT as their end-to-end asset management system, integrated with Ellipse and the site specific systems.
- High-level agreement of the AMT modules/functionality that both sites would use, albeit independently.
- High-level agreement of "To Be" processes, and data flows/integration between AMT and other site-specific systems.
- Detailed deployment, project scope and phases for AP in phases:
    a. Phase 1 encompassed upgrading their existing AMT software and deploying integration to Ellipse, and
    b. Phase 2 was the rollout out of AMT for full work management, with additional integration to Ellipse and other site systems

In March 2019, a workshop was held at LB in order to define a detailed scope for the AMT implementation project. LB confirmed it wanted to use AMT per the decisions made in September 2018. However, at this time, additional integration, training and change management was requested. Accordingly, RPM submitted a proposal to LB for the implementation project.

AP's Phase 1 project commenced in April 2019.

By July 2019, AP had almost completed Phase 1, and in parallel, had started their corporate Glencore Copper project to replace Ellipse with SAP by 1st January 2020, without Orbfusion (at Glencore Copper Africa AMT integrates with Orbfusion, and Orbfusion manually integrates with SAP).

LB is already live with their corporate Glencore Copper SAP environment. They made a decision to switch off their Ellipse environment, but without commencing their AMT deployment project, they are currently managing maintenance manually. As a short-term workaround, LB has opened in SAP numerous future dated work orders to capture material movements and costs, and is using Excel for planning and scheduling maintenance activities.

In August 2019, Glencore Copper South America's management team approached RPM with the idea of LB and AP using a single, common, standard AMT platform – same software, same configuration, same functionality, integration, same IT architecture/hardware, etc. Additionally, Glencore requested that the platform be in place by 1st January 2020. RPM advised that a workshop would be required to determine whether (a) a single, standard solution is possible, and (b) whether it could be implemented by 1st January 2020. The workshop was held in Santiago on 6th – 8th of August 2019. It was attended by Glencore and RPM, and this document serves to answer those questions.

1.2. Why a single instance?

Glencore's goals for a single instance of AMT supporting Antapaccay ("AP") and Lomas Bayas ("LB") are:

- To drive a standard asset management framework for Copper South America with respect to systems, data, processes and KPIs.
- To be able to benchmark asset performance across sites as well as within each site.
- To avoid the management of two systems.
- To have symmetry with SAP, i.e. one SAP and one AMT.

With the implentation project of a single SAP instance driving the opportunity to implement standards and alignment at LB and AP across financials and materials management, Glencore identified a single instance of AMT as being anopportunity to help drive standards and alignment across asset management.

1.2.1. RPM Single Instance Feedback

Following on from the workshop, and subsequent investigation, it is possible to run AP and LB as two separate organisations within a single 'instance' of AMT as 'branches'.

However, there would be some limitations and restrictions.

- Both organizations would operate within the one time zone as set at the application / database server level (see section 3.5). Chile also has Daylight Saving time for part of the year and would mean a time variance for part of the year.
- Both organizations need to agree on the alignment of master data, and the ongoing processes that will manage and maintain this alignment.
- Both organizations need to agree what language to use for common descriptions.
- Alignment of asset numbering, identification and nameplate information must be decided.
- Reporting requirement alignment.

There may be additional items identified as the solution workshops further refine system configuration and use across both organisations. RPM does currently have some customers operating across time zones in a single instance of AMT and while time zone issues appear not to have been significant for them, they do not operate and manage fixed plant assets, and their solution design is not identical to LB and AP.

1.3. Why Go-live at 1st January 2020?

Glencore's motivation for an AMT go live on 1st January 2020 is to align with SAP Go-live at AP, because when SAP goes live at AP, AP's Ellipse will be switched off. If AMT is not live when Ellipse is switched off, then AP would be without a maintenance planning and scheduling system. AP would have to resort to spread sheets and other disconnected/manual methods of running their maintenance operations. This is how LB is operating at present – SAP is live, and AMT is not – as a result LB's maintenance operations are using manual systems to manage its maintenance operations.

1.3.1. RPM Go-live Feedback

RPM's feedback is that Go-live by 1st January 2020 is not a realistic timeframe. There are many reasons for this.

A typical AMT deployment project is 6 months and is characterised by:

- Single company/operation
- No requirement to standardise and convert data across multiple operations

- No requirement to standard asset management processes across multiple operations
- Deployment of RPM's standard SAP adaptors

The AMT deployment project for AP and LB is not a typical AMT deployment project. For example,

- The scope is larger (e.g. multiple companies/operations – AP and LB).
- There are non-typical solution requirements (e.g. single AMT instance for both companies to use).
- A lot of ground-work is required to meet Glencore's standardisation goals (e.g. complete alignment of system table information, complete transformation of AP and LB information into a single combined set of information which involves standardizing and converting data; standardizing maintenance processes).
- Integration has to be developed (Glencore has to build integration to AMT's adaptors rather than using RPM's standard SAP adaptors due to the nature of Glencore Copper's SAP installation). This is a significant risk to the project.
- There is an SAP deployment project happening in parallel (which could absorb Glencore project resources, could make AMT trainees unavailable; and can delay AMT project steps that require SAP, e.g. SAP data needed for AMT system table standardization/set up/conversion, SAP available for integration testing).

Consequently, RPM strongly believes a 1st Janaury 2020 date for Go-live is not achievable, and rushing the implementation for two sites will ultimately deliver a sub-optimal solution and compromise solution adoption.

2. AMT Solution Design

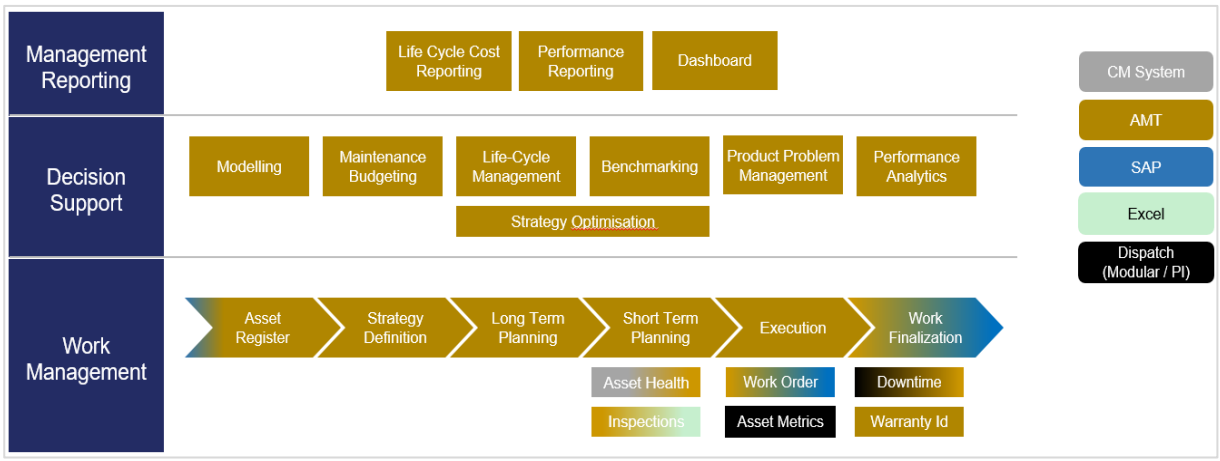
Normally RPM would deploy AMT-SAP integration using RPM's standard SAP adaptors, which are based on RPM's SAP certified architecture. However, due to the nature of the SAP installation at Glencore Copper Africa, deployment of those adaptors is not possible. As a consequence the solution design for AMT at AP and LB project, presented in this document, is functionally and technically constrained.

There are three broad asset management processes supported by AMT. These are:

- Work Management
- Decision Support
- Management Reporting

Figure 1 illustrates the processes supported by AMT.

Figure 1 AMT Solution and Functions



The following AMT modules/functionality are excluded from the Solution proposed for AP and LB:

- AMT Mobile – Timesheets
- Advanced Condition Monitoring (CBM)
- Piece part-level Warranty Identification
- Core Workflow
- Functional Locations

AMT-IC report writing module for customer developed reports is included.

2.1. Work Management Processes

An AMT deployment supports the following Work Management processes:

2.1.1. Asset Registration

Glencore has decided that AMT will be the master of the Asset Register for maintainable assets. Therefore, the following statements are acknowledged:

- Assets will be independently registered, and terminated, in AMT manually by a Central Administrator.
- Assets cannot be interfaced from SAP to AMT. As a result, SAP may not identify all maintainable assets.

- AMT will provide to SAP's data stewards any new AMT asset data for manual update in SAP.
    - Glencore's asset registration form, which is provided to the AMT system administrator to enter the asset into AMT, will also be provided to the SAP data steward so the same asset can be created in SAP.
- Note: Glencore acknowledges that functional locations may not be needed in AMT and is therefore currently excluded from the solution design (AMT's functional locations functionality is not deployed at Glencore Copper Africa for example). The final outcome for LB and AP depends on further discussions with Glencore's SAP team during detailed integration and solution design workshops.
- Glencore is responsible for the reconciliation between asset data held in AMT and SAP.
    - RPMGlobal is not responsible for the resolution of any inconsistencies at the record level between AMT and SAP. AMT is the master record and SAP will need to be adjusted accordingly.
- Forecast usage assumptions for an asset will be maintained in AMT.
- Automatic updating of usage forecast will be disabled in AMT

2.1.2. Strategy Definition

- AMT is the master for strategy tasks and standard jobs for assets managed by AMT.
- Standard jobs are created and maintained in AMT.
- Parts, parts pricing and part supersessions are imported into AMT from SAP using AMT's Parts Adaptor.
- Strategy tasks are created and maintained in AMT.
- Electronic workscopes (required by AMT Mobile) are created and maintained in AMT.
- Other job instructions will be linked to AMT Strategy Tasks via Glencore's SharePoint software, which will be available at both sites.
- Strategy Tasks (maintenance plans) and Standard Jobs (task lists) will not be maintained in SAP.

2.1.3. Long-term Planning

- Long-term planning is performed in AMT.
- AMT will produce a component forecast report.
- AMT will produce the piece parts forecast report based on standard jobs linked to strategy tasks.
- AMT will produce the availability forecast report for mobile mining equipment.
- AMT wil produce a labour forecast report.

2.1.4. Asset Health

- AMT Mobile will be used to capture visual condition monitoring measurements.
- Notifications raised in AMT, as a result of condition monitoring measurements recorded in AMT Mobile, will be reviewed and actioned by Supervisors in AMT Client.
- Glencore have agreed that no integration to condition monitoring systems, and advanced condition monitoring (i.e. CBM functionality) is included, or expected, in this implementation project.

2.1.5. Inspections

- AMT Mobile Inspections will be deployed.
- It has been agreed that AMT Mobile Inspections will be deployed. In the first instance, this will be for a specific Work Group and a specific Process Plant Area at one, or both, of Antapaccay and Lomas Bayas.
- Measurements relevant for AMT Mobile Inspections will be loaded into AMT prior to go-live.
- Notifications, generated by AMT Mobile Inspections measurements, will be reviewed and actioned by Supervisors in AMT Client.

2.1.6. Short-term Planning

Glencore have agreed to consistent AMT planning and scheduling processes across LB and AP.

Planning

AMT is the master work order system.

- AMT will automatically generate maintainable asset work orders from strategy tasks.
- Notifications, if required, will create AMT work orders.
- Other backlog reactive and repair jobs create AMT work orders.
- AMT will generate a work order number for all AMT work orders.
- For a work order linked to a standard job via a strategy task or not, the standard job parts are added to the AMT work order automatically.
    - Users can modify these on the AMT work order if required.
    - Glencore acknowledges that AMT Inventory Search is not included at this stage due to integration restrictions applied by Glencore Copper SAP configuration.
- Work order authorisation occurs in AMT. AMT configuration supports work order authorisation based on value.
- AMT will generate work orders into SAP using AMT's work order adaptor.
    - Based on AMT's work order, SAP will create a single SAP work order corresponding to each AMT work order.
    - AMT's work order number will be stored in a reference field of the corresponding SAP work order.
    - Glencore have advised that only work orders requiring parts and misc. items will be opened in SAP.
        - Glencore have advised that Misc Cost items are treated as parts in SAP.
    - Glencore have advised that labour costs are not charged to SAP work orders. Labour is treated as an overhead. Therefore, any labour only work orders in AMT are not required to be integrated with SAP work orders.
    - The work order creation message from AMT will contain a Branch Code that will enable SAP to distinguish into which SAP Company the work order is generated (Lomas Bayas or Antapaccay).
- Planned parts sent from AMT are added to the SAP work order using AMT's work order adaptor as Planned parts.
    - If AMT sends planned parts updates for the same work order to SAP, SAP will adjust the existing SAP work order; SAP will not open an additional work order.
- In SAP, Parts are issued from stores to the work order. Parts requisitions and any other requisitions are managed in SAP.
- SAP returns the Job Ready Status of parts to the AMT work order.

Scheduling

- Users schedule AMT work orders.
- Work order scheduling, at the work group level, will be performed in AMT. Available labour hours will be manually defined in AMT per work group; available hours will not be imported from an external system).
- Eventscopes and work scopes (the Job Package) printed from AMT by the Planner-Schedulers, are provided to workshop/field supervisors.

2.1.7. Downtime

- Actual downtime is captured in Modular Mining ("MM") for mobile mining assets, and in PI/Historian ("PI") for process plant assets.
- MM and PI exports actual equipment maintenance downtime in real-time to AMT.
    - Note, Glencore have agreed that AMT will only import 5 critical process plant productive units for PI across each of LB and AP.

- Downtime notifications imported into AMT will be placed into a holding area. Users will review, and decide whether to create an AMT Event from each downtime notification.
- Actual uptime for the same Event will be captured in Modular/PI, and will be exported (via interface) to AMT.
- Users will be able to see the status of equipment (maintenance completed, maintenance in progress, upcoming planned maintenance events).

2.1.8. Work Execution

- Relevant users (e.g. supervisors, technicians) with access to AMT Mobile, can
  - Complete AMT work orders in AMT Mobile, and enter work order feedback into AMT Mobile work orders
  - Capture new defects in AMT Mobile
  - Capture actual usages in AMT Mobile
- Relevant users (e.g. supervisors, technicians) without access to AMT Mobile, will complete other job folder documentation as per the normal process on site manually, using AMT Eventscope and AMT workscope documents. Clerks, using this documentation and AMT Client, will:
  - Complete AMT work orders in AMT, and enter work order feedback into AMT
  - Enter defects in AMT

2.1.9. Work Finalization

- Any component serial number movements (i.e. component serial number installed and component serial number removed) are manually recorded on an AMT work order.
- Work Order Parts & Misc.:
  - Glencore will perform all actual actions on work order Part lines (“materials”) and Misc. lines in SAP.
  - Actual parts and misc. costs for a work order will be imported from the SAP work order onto the AMT work order.
- Work Order Labour:
  - Actual labour hours will be captured manually by clerks on AMT work orders.
  - Actual labour costs will not be captured on AMT work orders or imported from SAP.
- AMT will not import actual overhead costs from SAP into AMT. It has been agreed by Glencore Antapaccay that this is a change in process. The impact of this change to the Budgeting process process is understood by Glencore.
- On a monthly basis, actual costs will be manually reconciled between AMT and SAP. This is due to SAP not necessarily having all maintainable assets registered.

2.1.10. Asset Metrics

- Glencore has agreed that actual usages (hours, meters, tonnes, etc) will be manually entered in AMT.
  - This decision can be revisited in future if actual usage data can be integrated from MM and PI.
  - Glencore is responsible for actioning and investigating data integration from MM and PI.

2.1.11. Warranty Identification

- AMT can flag component warranty on the AMT work order based on the warranty period entered on the corresponding AMT strategy task.
- LB and AP will track warranty claims in AMT using AMT work orders.

2.2. Decision Support Processes

The following decision support processes will be supported by AMT:

2.2.1. Strategy Optimization

- Out-of-the-box AMT strategy optimization functionality will be used, including Strategy Task Analyser, to identify strategy optimization opportunities, and develop and compare “What If” alternate maintenance strategy options
- AMT can create What If scenarios.
- Maintenance strategy changes are made directly in AMT.
    - Glencore have agreed that maintenance plans and task lists will not be created in SAP for either LB or AP and no information passed from AMT to SAP. There are no integration requirements.

2.2.2. Performance Analytics

- Using feedback information captured on the AMT work order during the Work Finalisation process, AMT users can analyse:
    - Component change out history (life achieved, change out cost).
    - Component performance (symptoms and causes of failure).
    - Maintenance costs (systems, subsystems, components, parts, etc.).
    - Downtime, downtime delays and repetitive stoppages (Pareto analysis).

2.2.3. Product Problem Management

- AMT will register and track the resolution of product problems following the AMT process.

2.2.4. Modelling

- AMT can model and compare the direct maintenance costs of new assets between different suppliers/OEMs which Glencore are considering to purchase.

2.2.5. Benchmarking

- AMT can run benchmark reports between assets on at LB or AP, and assets across both LB and AP to compare:
    - Cost per Hour of a fleet of trucks on a specific site; and
    - Cost per Hour of a model of truck across LB and AP sites

2.2.6. Life-cycle Management

- AMT’s standard Residual Value in Component (RVIC) report models the economic life of assets

2.2.7. Maintenance Budgeting

- AMT will generate Glencore’s direct asset cost budgets, excluding direct maintenance labour costs for assets managed in AMT.
- Glencore have agreed that any budgets developed in AMT will not be exported to an external corporate budgeting system
- Antapaccay’s indirect/overhead cost budgeting process currently supported by AMT will be discontinued.
- Glencore have agreed that neither LB or AP need to budget indirect/overhead costs in AMT.

2.3. Management Reporting

AMT will provide the current standard management reports for the following:

2.3.1. Life-cycle Cost Reporting

- AMT reports can analyse the equipment life cycle cost (historical costs and future costs).

2.3.2. Performance Reporting

- AMT has standard industry reports, including
  - Physical Availability
  - Mechanical Availability
  - MTBF
  - MTBS
  - MTTR
  - Service Accuracy
  - Planned vs. Unplanned Events
  - Maintenance Compliance
- Glencore have agreed that any variances to these standard reports will need to specified, and addressed by custom developed reports.
  - Glencore Copper is introducing standard metrics for its sites.
  - Below is RPM's feedback with a cavaet with respect to support of these metrics:
    - RPM notes there are several methodologies on which these reports can be based.
    - Without Glencore providing their actual report specifications it is not possible to categorically state complete compliance with standard reports and/or with the standard reporting cubes from which customised reports can be written.

Table 1 AMT Reporting Feedback

Metric	RPM Feedback
Overdue Statutory Work	An AMT work order can be flagged as Statutory, and can be reported as overdue.
Schedule Loading	AMT can store by work group (i.e. work centre) available hours. An AMT work order holds planned labour hours by work group. Both can be compared.
Schedule Compliance	AMT is able to report % of planned work orders actually completed.
PM Work Not Executed	Assumed to be a subset of Scheduled Compliance, focussing on PM only work orders.
PM Work > 7 Days Overdue	Each PM work order has a Target Date and if scheduled a Planned date, which can be reported in terms of > "x" days overdue.
Effective Resource Utilisation	Actual Labour Hours per work group can be recorded on AMT work orders, which can be compared to Available hours per work group.
Planning Accuracy - Labour	AMT is able to compare Planned labour hours on work orders vs. Actual labour hours on work orders; for work orders which were planned (not breakdown).

- The AMT-IC module will be deployed as part of the solution. This enables Glencore to write their own reports using AMT's Live Views and Reporting Cubes, where AMT's standard reports are not suitable or the report is not available out-of-the-box.

2.4. System Integration

The integration for this solution will not be delivered as a standard SAP integration using RPM's SAP adaptors, due to the nature of the SAP installation at Glencore Copper Africa. Instead, the approach for this solution will be for Glencore (or a 3rd party appointed by Glencore) to develop the EPF-SAP side of the integration utilising AMT's standard adaptors. This forms a critical path of the deployment project and puts the project's timeline for Go-live and solution at risk.

- Glencore's Responsibilities:
  - Glencore (either itself or by appointing a 3rd party directly) is responsible for designing, developing, testing, deploying, maintaining and supporting the EPF-SAP side of the interface. Specifically, the following interfaces:
    - Parts (SAP to AMT)
    - Parts Pricing (SAP to AMT)
    - Supercede Parts (SAP to AMT)
    - Create SAP Work Order (AMT to SAP)
    - Update SAP Work Order – planned parts (AMT to SAP)
    - Parts Ready Status (SAP to AMT)
    - Work Order Financial Summary (SAP to AMT)
    - Actual Work Order Details (SAP to AMT)
  - User Acceptance Testing will be managed by Glencore with support from RPM and Glencore.
- RPM Responsibilities:
  - RPM will assist Glencore with the validation of their integration interfaces and will unit test the AMT side of the integration by confirming the integration as and between AMT and SAP, AMT and MM, and AMT and PI.
  - RPM will assist Glencore create test scripts.
- Both RPM and Glencore agree that any interfaces which are not developed, tested and signed off by both RPM and Glencore prior to the official Go-Live date will be excluded from the scope of the project and therefore excluded from final project sign off. The cost to complete the outstanding interfaces will be determined using RPM standard consulting rates and performed as a separate project.

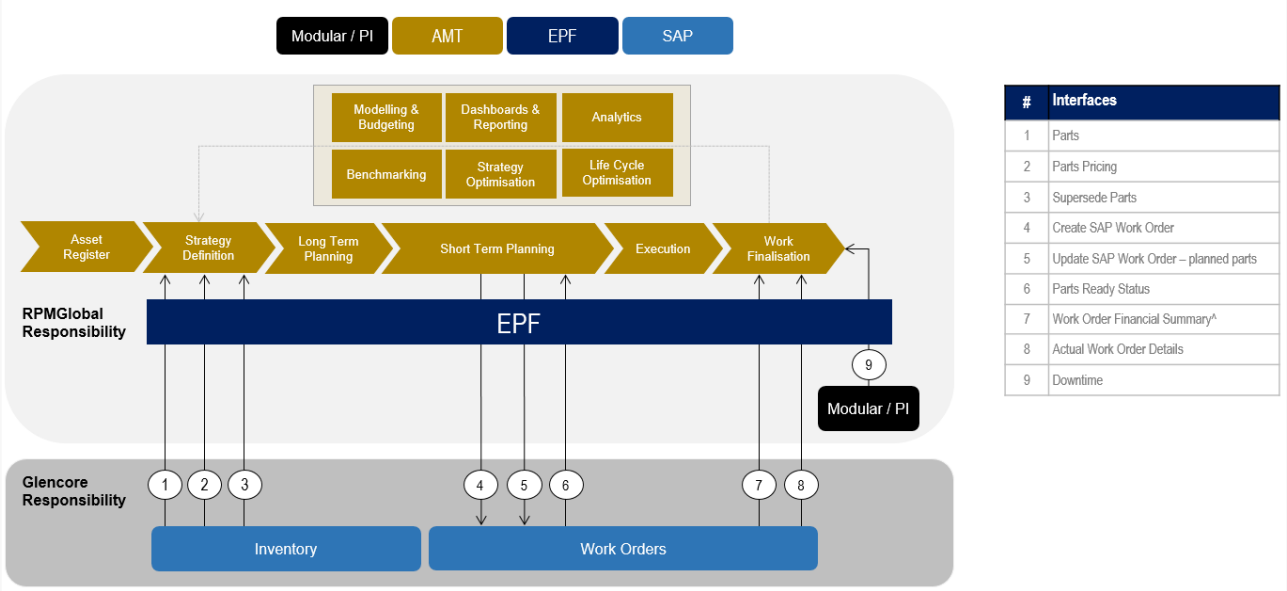
During the August 2019 workshop, Glencore were not aware of any SAP configuration differences between LB and AP which would prevent AMT from being deployed on a single platform. However, no SAP experts attended the workshop.

Critically, an Integration Design workshop with Glencore will be required in order to confirm integration point data flows and Glencore's ability to deliver the EPF-SAP side of the integration for this solution. All of which may impact and change the overall AMT solution, configuration, and consequently the ability for LB and AP to operate on a single, standard AMT platform.

2.4.1. Integration Data Flows and Descriptions

Figure 2 depicts the desired Integration Solution Design for AMT at LB and AP. Table 2 summarises the individual interface descriptions.

Figure 2 AMT Integration Solution Design Framework



#	Interfaces
1	Parts
2	Parts Pricing
3	Supersede Parts
4	Create SAP Work Order
5	Update SAP Work Order – planned parts
6	Parts Ready Status
7	Work Order Financial Summary^
8	Actual Work Order Details
9	Downtime

Table 2 Interface Descriptions

#	Adaptor Name	Description
1	Parts	• Parts are imported from SAP; AMT users use these parts when creating standard jobs in AMT.
2	Parts Pricing	• Part pricing is imported from SAP into AMT.
3	Supercede Parts	• Part supersessions are imported from SAP into AMT.
4	Create SAP Work Order	• Work orders are generated by AMT • AMT will send a message to SAP to create the SAP work order, the message includes an AMT Branch/Site code enabling SAP to identify whether the work order belongs to LB or AP (an SAP Company). • Parts sent from AMT to SAP will be added to the SAP work order as planned parts. • This process creates a demand on SAP Inventory; it does not action an SAP Inventory issue.
5	Update SAP Work Order – planned parts	• Users can add/remove additional planned parts on the AMT work order. These changes are sent to SAP and update the planned parts on the existing SAP work order. This process will not create additional SAP work orders.
6	Parts Ready Status	• SAP sends back to AMT periodically the Job Ready Status at the part level.
7	Work Order Financial Summary	• Summaries of actual costs of work orders (parts and miscellaneous) are sent from SAP to AMT (labour is not costed on the SAP work order). ^ This interface may/may not be required depending on the capability of SAP with respect to interface #9.
8	Actual Work Order Details	• Actual work order details will be imported into AMT from SAP.

#	Adaptor Name	Description
9	Downtime	• Actual down and uptimes are sent from MM to AMT for mobile equipment, and from PI to AMT for 5 critical process plant productive units for LB and 5 critical process plant productive units for AP.

- Note, the AMT interface which performs a real-time inventory look-up from AMT to SAP (or via intermediate repositories like Orbfusion) to search for parts and view stock on hand, has been excluded from the scope due to RPM's understanding of integration limitations of the nature of the SAP installation used by AP and LB.

2.4.2. Standardisation

Glencore is responsible to ensure uniqueness in asset identification/numbering and work order numbering in SAP across both LB and AP. Glencore are responsible for the resolution of any conflict in asset identification/numbering across LB and AP and between AMT and SAP. AMT being the master will indicate that an SAP duplication will need to be resolved.

Glencore is responsible for standardizing unit of measure codes, downtime codes and asset numbering between Modular Mining and PI/Historian across both LB and AP. This will ensure AMT has a single set of downtime codes for reporting, and will ensure AMT users do not become confused as to which downtime codes to use, and the meaning of the downtime codes.

It has been agreed that EPF will not maintain any code translations.

2.5. AMT System Data Entities

The main data entities and a description of how each will be maintained are detailed in Table 3.

Table 3 System Data Entities

Item	Description
Users	Maintained in AMT manually
User Roles	Maintained in AMT manually
Asset	Maintained in AMT manually
Work Groups and Employees	Maintained in AMT manually
Strategy Tasks and Standard Jobs	Maintained in AMT manually
Usage Profiles	Maintained in AMT manually
Actual Usages	Maintained in AMT manually
Eventscopes and Workscopes	Generated from AMT
Workscopes – Attached documents	Link to a single Sharepoint document management system used by both LB and AP
Workscopes – Electronic	Created in AMT (to be used by AMT Mobile Inspection App) manually
Work Order Settlements	Imported from SAP
Events/Downtime	Imported from MM and PI
Measurements	Captured via AMT Mobile
Measurement Rules	Maintained in AMT manually
Notifications	Sourced from AMT Mobile, and reviewed and actioned in AMT Client
Budgeting	Detailed direct asset maintenance cost forecasts developed in AMT (excluding labour costs)

2.6. AMT Software Enhancements

- None identified.
- Any software modifications will be treated as a variation to the contract.

3. Solution Configuration

In order for a single LB and AP to utilize a single AMT Production instance/platform, a standard configuration of AMT is required.

The sections below detail the standard areas for configuration.

3.1. AMT Functionality

AMT will be installed with English and Spanish language packs.

The following functionality is to be deployed:

Table 4 Functionality to be Deployed

Functionality	Description
Modelling	• Equipment purchase evaluation • Calculate equipment Life Cycle Costings • Fleet modelling
DLCC & Budgeting (Dynamic Life-Cycle Costing)	• Life-Cycle Costing and analytics • Forecasting and tracking of equipment maintenance costs • Strategy definition (including standard jobs) • Strategy optimisation • What-If analysis • Asset economic life optimization • Long-term planning • Direct equipment cost budgeting
Scheduling & Shiftlog	• Backlog management • Work order planning • Work order scheduling • Shiftlog • Downtime Capture • Component Tracking • Equipment Performance KPI reporting • Maintenance Efficiency KPI reporting
Health	Standard Condition Monitoring • Analyse historical failures • Capture condition monitoring data • Interpret condition monitoring data against rule sets to trigger Notifications • Notification Management
Warranty Identification	• Ability to identify potential component warranty opportunities
AMT Mobile	Field users (eg. mechanics, operators, inspectors, supervisors) • Inspections • Defect capture • Usages capture • Close Work Orders
Dashboard	• Customisable for all users (mobile or non-mobile)
AMT-IC	• BI tool to write custom AMT reports using prebuilt data cubes and views

3.2. AMT User Security-level Configuration

Glencore accepted the AMT current design and security permissions (and the associated limitations) across the processes of asset and work management, strategy management and optimisation, budgeting and forecasting and reporting.

Glencore has agreed that any Central System Administrator can access and modify system information at both sites.

3.2.1. Groups

- Groups will be set up in AMT per User Role, with specific permissions
- User Roles will be site specific
- This translates to different Groups for LB and AP
- Group and group permissions will be administered by a Central System Administrator

3.2.2. User Accounts

- User Accounts will be administered by Central System Administrator
- Users will be associated with a:
    - Group
    - Language
    - Branch / Site / Equipment access
    - Work order authorisation level

3.2.3. Global Data Visibility and Access

The majority of the data which functionality in AMT accesses, is governed by the Branch/Sites to which a user is given permission. However, there are some functions which have global data access. In the August 2019 workshop this was explained to Glencore, and a specific example given: AMT's Budget and Forecast Manager. If a user has access to this function, then the user is able to access and alter any budget; regardless of whether the user is restricted to only LB or AP sites.

Allowing global access in some areas of AMT is deliberate, and by design, since AMT is designed to be used by a single company.

Glencore accepted that there will be areas within AMT that both sites will be able to access, review and modify each other's data. Glencore acknowledged the current design of AMT in this regard, and that this reality would be accepted so that LB and AP did not have to install separate instances of AMT.

All global configuration points will affect both operations; these will be highlighted and understood during the initial implementation project and reiterated throughout the project.

After Go-Live, continued Glencore project governance and change management processes will need to keep the operational alignment for single instance configuration.

3.3. AMT System-level Configuration

AMT has many global configuration settings which apply to all AMT branches, sites, fleets and assets, and AMT functionality, security and integration. These settings can have a direct impact on whether a single AMT instance/platform for LB and AP is achievable. Meaning, if LB and AP have a different requirement, which necessitates a different system-level configuration, then a separate AMT instance will be required (i.e. a single platform for both sites will not be possible).

During the August 2019 workshop, many (but not all) of the global configuration settings were discussed, in particular those which related to supporting a single platform for LB and AP. The following AMT system-level configuration decisions for the standard solution are summarised in Table 5.

Table 5 AMT Configuration

Configuration	Decision
Functional Locations	• Disabled.
Equipment Hierarchy - Branch	• Branch will represent an Operation, i.e. LB and AP
Equipment Hierarchy - Site	• Site will represent a department within the Operation, i.e. LB-ProcessPlant, LB-Mining
Equipment Hierarchy - Fleet	• Fleet will represent an Area/Process for Fixed Plant departments, and will represent actual fleets for departments with mobile equipment
Currency	• AMT will be configured with a single currency which will be US Dollars (USD) • All reporting will be in USD • All transactions imported into AMT will be in USD
Auto-update utilization	• Auto-update utilisation will be disabled.
Number of strategy task work orders which can be opened in advance	• Both LB and AP have agreed that only 1 strategy task work order will be opened in advance.
Non-planning task rescheduling	• Non-planning task rescheduling tasks will be enabled • Non-planning task rescheduling factor = 33%
External Document location	• Both LB and AP will use the same instance of Sharepoint, and this will be the only external document management system referenced by AMT
Maximum attachment size	• Will be confirmed during deployment based on line-speed performance. Notwithstanding, both sites committed to appying the same rules. • RPM's recommended maximum attachment size is 200kb.
Auto Cost Accrual & Negative Accrual	• Cost accrual relates to accuring cost for work-in-progress strategy task work orders. Negative accrual relates to future-dated work orders with actual costs. • Will be confirmed with Finance during deployment. Notwithstanding, both sites committed to appying the same rules.
Budget RVIC calculation	• This relates to which strategy tasks should be included in AMT's budgeting module's RVIC calculation. • Will be confirmed during deployment. Notwithstanding, both sites committed to appying the same rules.
Overheads	• Budgeting and/or capturing overhead costs is excluded from the solution. Both sites have committed to this exclusion.
Advanced Asset Health	• Excluded from the solution.
AMT Work Order Mode	• "PO Create" Mode will be enabled, which enables a user to pass multiple AMT work order planned parts updates to SAP.
Auto-Create Work Order Number Mode	• Due to AMT being the work order system, AMT will be configured so that AMT generates a work order number for

Configuration	Decision
	work orders created in AMT. AMT's work order number will need to be stored on the SAP work order, in order for SAP work orders to link to AMT work orders.
AMT Work Order Authorisation	• Advanced Authorisation Mode
Capture of actual labour hours	• Both sites will capture actual work order labour hours on AMT work orders manually in AMT Client. Labour hours will not be imported from SAP or recorded using the AMT Mobile – Timesheets process.
AMT Mobile - Timesheets	• Excluded from the solution
External Downtime	• Both sites will import downtime from external systems. • Downtime notifications imported into AMT into a holding area. Users will review, and decide whether to create an AMT Event from each downtime notification.
Quick Watch Horizon	• This related to how many days in the past Quick Watch should consider for certain Quick Watch Views. • Will be confirmed during deployment. Notwithstanding, both sites committed to appying the same rules.
Piece Part Warranty Identification	• Excluded from the solution.
Financial Periods	• For both operations, the Financial Periods are the same where for example the 1st period of a financial year equals the full month of January, the 2nd period of the financial year equals the full month of February and so on.
Budget Calendar	• For both operations, the annual budget calendar is the same as the standard Gregorian calendar
Time Zones	• AMT will use the time zone of its server.

3.4. AMT System Table-level Configuration

AMT contains a number of system or reference tables. As part of an AMT deployment, these system tables need to be defined by the operation installing the software. However, given this Solution is to be used by two Operations (LB and AP) then both LB and AP have to jointly define/standardise AMT's System Tables.

Below is the list of AMT system tables for the current release of AMT, as of the date of this document, which are relevant for the Solution. Both LB and AP need to standardise the contents of each system table, and ensure there is a standard definition as to its purpose.

Some considerationa are:

- Branch/Site/Fleet system tables will be dictated by RPM based on a combination of AMT security and integration considerations, and Glencore's desired to have a single platform for both LB and AP
- Some system tables will have its purpose and contents dictated by SAP (e.g. cost centres, expense elements)
- Some system tables will have the flexibility to accommodate operation specific data (e.g. the Employees system table would contain employees from both sites)

The Integration column indicates if a system table is potentially to be influenced by, or used during, integration with external systems (e.g. SAP, MM, PI).

- If a system table is designed to be dictated by, or used during, integration (determined by the Integration Design Workshop), then values populated into AMT's system tables will need to be standardised across LB and AP.
- Otherwise, there is risk integration may fail and/or users of AMT will be confused if they see two or more codes in AMT for the same thing or the same code in AMT actually meaning two different things (e.g. Modular classifies maintenance downtime as "1" but PI classifies operational downtime as "1".

The Classification column indicates whether a system table will typically have direct input from the operation "Operation" as opposed to being defined as a "Standard", by a central standards committee for all operations.

- It is important to note that the majority of AMT's system tables can be viewed by all users, regardless of whether the user is located at LB or AP. For example, even though a user is located at LB, he/she can create a new equipment and assign it to an AP Cost Centre. Glencore recognises this is how AMT functions and has accepted it.

Table 6 System Data Entities

System Table Grouping	System Table	Integration	Classification
Organisation	Region		Standard
Organisation	Division		Standard
Organisation	Branch Administration	Y	Operation
Organisation	Site Administration	Y	Operation
Organisation	Fleet Administration	Y	Operation
Organisation	Industry		Standard
Organisation	Industry Group		Standard
Organisation	Work Location		Operation
Organisation	Work Group	Y	Operation
Organisation	Employee	Y	Operation
Financial	Cost Bearer	Y	Standard
Financial	Cost Centre	Y	Operation
Financial	Expense Element	Y	Operation
Financial	Cost Responsibility		Operation
Financial	Currency	Y	Standard
Financial	Escalation		Operation
Financial	Financial Periods	Y	Standard
Maintenance	Component Code Hierarchy		Standard
Maintenance	Modifier Code		Standard
Maintenance	Task Type		Standard
Maintenance	Task Counter		Standard
Maintenance	Job Code		Standard
Maintenance	Occurrence Type		Standard
Maintenance	Maintenance Type		Standard
Maintenance	Operational Criticality		Standard

System Table Grouping	System Table	Integration	Classification
Maintenance	Priority		Standard
Maintenance	Symptom		Standard
Maintenance	Cause		Standard
Maintenance	Event Prevention		Standard
Maintenance	Source		Standard
Maintenance	Task Planned Time		Standard
Maintenance	Responsibility	Y	Standard
Maintenance	Activity	Y	Standard
Maintenance	Variance Cause	Y	Standard
Maintenance	Service Codes		Standard
Maintenance	Issue Type		Standard
Maintenance	Supplier	Y	Standard
Maintenance	Scheduling Group		Standard
Maintenance	Task Misc Category		Standard
Parts and Standard Jobs	Labour Activity		Standard
Parts and Standard Jobs	Part Type	Y	Standard
Parts and Standard Jobs	Parts Price Group	Y	Operation
Parts and Standard Jobs	Source of supply	Y	Standard
Parts and Standard Jobs	Parts Classification		Standard
Equipment	Equipment Class		Standard
Equipment	Equipment Criticality		Standard
Equipment	Equipment Group		Standard
Equipment	Equipment Group – Registration		Standard
Equipment	Equipment Group – Configuration		Standard
Equipment	Equipment Group - Application		Standard
Equipment	Manufacturer		Standard
Equipment	Model		Standard
Equipment	Equipment Category		Standard
Equipment	Equipment Location		Standard
Equipment	Equipment Status		Standard
Usages and Measurements	Units of Measure	Y	Standard
Usages and Measurements	Measurement Code	Y	Standard
Other	Document Type		Standard

3.5. IT Architecture

Based on the assumption of a single AMT instance for both LB and AP, the following environments will be required:

1. Production server
2. Training server
3. Test server
4. Development server

All AMT environments will be accessible for LB, AP and RPM. It is RPM's understanding that the server will be located in Santiago, and therefore the server date and time setting will be Chile Standard Time and Chile Summer Time for daylight saving.

3.5.1. Time Zone

AMT uses the server time to record timestamp data as information is created, captured and stored. As both Antapaccay and and Lomas Bayas will be using a single server instance then the time will be based on the configuration of the server time. There is a one or two hour time difference between the operations depending on the time of year and Chile's daylight saving time.

Therefore if the server is located in Santiago, the date and time information for records created for Antapaccay will be one hour ahead of the actual date and time.

3.5.2. Time Zone Impact

Items recorded in the AMT database via integration have created date and time stamps that are equivalent to the AMT server date and time.

Depending on the user, exposure to time zone impacts may be limited visually on screen. For example, in the scheduling screen the current timeline corresponds to the local desktop settings. Retrieving transactions, an occurrence for example, will show the date saved with the transaction. This means that an occurrence dated 11 pm on 21-Aug-2019 will show with that date whether in Peru or Chile.

However, there are known inconsistencies with time and date behaviours on the work order entity table as detailed in Table 7. The transaction, created and approved in Antapaccay at 10:00 am with an immediate demand for parts is running off server time (11:00 am). The parts available date is set as one hour later than the actual perform by date and time.

Table 7 System Data Entities

Action	Time Zone Setting	Example Time
Create work order	Server setting	11:00 am
Notification Date (for authorisation)	Server setting	11:00 am
Last Modified Date	Server setting	11:00 am
Parts Available Date	Server Setting	11:00 am
Perform By Date	Desktop setting	10:00 am
Authorised Date	Server setting	11:00 am
YTS update – Planned Start Date	Desktop setting	10:00 am
Updated Last Modified Date	Server setting	11:00 am
Updated Parts Available Date	Server setting	11:00 am
Planned Down Time	Desktop setting	10:00 am
Completed Work Order	Server setting	11:00 am

These date and time issues do not manifest themselves when operations are in the same time zones. Nor experienced by our existing customers that operate across multiple time zones due to the AMT functionality they have implemented and use.

It is expected that fixed plant maintenance will need management at the in-shift level e.g. hourly. This raises a reporting risk for schedule loading and schedule compliance reporting at work group level.

For actual downtime captured in Modular Mining for mobile assets, and in PI/Historian for process plant assets, the information could potentially be one hour out of sync for Antapaccay depending on integration. Integration is real-time. Therefore, the created date and time for Lomas Bayas will be the same

as the downtime date and time. For Antapaccay, the date and time for the downtime will potentially be one hour before the created date and time.

Any AMT alerts and messages are raised with the desktop time zone settings but recorded with the server time zone setting. A reminder set on the same message by the user where the user is asked to select a date, is saved to the server with a transaction time as well as date that corresponds to the desktop date and time settings.

Not all organisations that currently operate with AMT across multiple time zones are performing schedule and execution processes in AMT. Most are managing AMT for mobile equipment via events where they place more importance on the event than the work items within the event. Their events are created in advance for repair and maintenance purposes. Any event information created manually uses the local desktop date and time.

Glencore is looking to manage both fixed and mobile maintainable assets. They will use events but not all work will be managed using events.

3.5.3. Impact of server setting for daylight saving decision

Chile is subject to daylight saving which causes another shift which, depending on the server setting, will either adjust the server dates and times for daylight savings or not.

If the choice is to adjust for daylight savings:

- LB will not be impacted.
- AP will experience a 2-hour difference.

If the choice is made not to adjust the server dates for daylight saving:

- LB will experience a 1-hour difference.
- AP will continue to experience their 1-hour difference.

4. Solution Deployment

The following sections outlines the key aspects of deploying the single instance of AMT for LB and AP.

4.1. Operations, Departments, Assets

- Antapaccay (AP) – two process plants and one mine, with approximately 1826 assets in total
- Lomas Bayas (LB) – one process plant and one mine, with approximately 1409 assets in total

4.2. IT Architecture

- A single AMT instance/platform will be configured for both LB and AP
- Glencore indicated that the environments will be set up in Santiago.
- A single AMT Test server and a single AMT Production server, and a single EPF Test and a single EPF Production server will be required which meets RPM's specifications for AMT and EPF
- If, during the implementation project, additional servers for UAT and Development are required, RPM will issue a variation for any additional effort required to install and configure these environments.
- The latest version of AMT and EPF will be installed on the respective servers

4.3. Data Preparation

- Glencore's central project team will define AMT's system tables
- Glencore's central project team will complete data templates supplied by RPM, populating the templates with the required data from LB and AP.
- Glencore is responsible for ensuring the data templates do not contain duplicate data.
- The following data from AP's existing AMT Production server will be extracted by RPM into Excel files and given to Glencore
    - Equipment registration details, excluding usage profiles
    - Current projection strategy tasks
    - Actual usages
    - Work order settlements
- Glencore is responsible for converting AP's information into the new standard solution codes and structures as defined by the Glencore central project team. They will then populate this data into the data templates supplied by RPM for upload into the newly configured AMT environment.
- The following data will be required for the new AMT solution
    - All system tables
    - Assets:
        - Equipment Hierarchy
        - Functional Locations (to be confirmed)
        - Assets
        - Current projection strategy tasks, with external document URL links to Sharepoint
    - Standard Jobs
        - Parts, parts prices
        - Standard jobs
    - Transaction history
        - 3 years of work order cost history
        - 3 years of actual usages history
    - Other
        - Electronic Workscopes for AMT Mobile (maximum 20 for LB and 50 for AP), including corresponding measurements and measurement translation rules
        - List of unique component serial numbers
        - Current backlogs which have yet to be created in SAP as work orders

4.4. User Training

- Approximate number of AMT trainees:
  - LB = 115, approximately 20 users will use AMT mobile only
  - AP = 110, approximately 40 users will use AMT mobile only
- Pre Go-live training approach:
  - RPM will conduct training for "key users", specifically Planners, Schedulers, Reliability Engineers, LTP, Budgeting (Cost Controllers at LB), AMT Champions, people who use IC, and Glencore trainers.
  - Train-the-Trainer (TTT) for lower end users specifically Supervisors, Clerks, General Managers, Dispatchers, Material Technicians, Superintendents
- Timing:
  - If the AMT user training schedule was to extend past Go-live, Glencore decided that in order to Go-live as close to 1st January 2020 as possible not all users will be trained Pre Go-live. Any users not trained Pre Go-live will receive on-the-job training and support from the Core project team and Glencore's trainers after Go-Live.
    - This approach is not recommended by RPM.
- Post Go-live training
  - RPM will be required to assist with Post Go-live on-the-job support at both sites
- Location:
  - Core project team and central administrators will receive training in Santiago
  - TTT delivered by RPM will be at a central location
  - Other non-TTT training delivered by RPM will be at each site

4.5. Standardisation

- Glencore is responsible for all standardisation activities, including but not limited to:
  - Data,
  - Reporting, and
  - Business processes which are relevant to enabling the single, standard AMT solution to be deployed and successfully adopted at LB and AP.

4.6. Change Management

- Glencore is responsible for any business process re-engineering activities and associated change management activities to redesign and/or implement new or enhanced business processes and procedures in preparation to roll out of the standard AMT solution.

4.7. Rollout Approach

- One operation will Go-live first, with all departments within the site going live at the same time.
- After completing Go-live at the first site, the project team will move to the second site for Go-live
- Which site goes first will depend on an assessment of their technical readiness in addition to their SAP readiness
- Any gap between each site going live is yet to be determined. It is anticipated that the gap would be, at a minimum, 1 month.
- With respect to AMT Mobile, AMT Mobile will only be rolled out to single work group within a single area at each site. Glencore will rollout AMT Mobile itself to other sites after the initial single site roll out.

4.8. Glencore Project Team

- Glencore will establish a dedicated, central project team who will be responsible for the AMT implementation project across LB and AP.
- A Glencore resource who is part of the central project team will have authority to issue directives and make key decisions with respect to AMT configuration settings, data, and system setup.
- There will be a central project manager, AMT champion(s), and system administrators.
- LB will provide 4 EFT dedicated resources to the project, AP will provide 3 EFT dedicated resources. There will also be 1 IT representative allocated to the project.

5. Project Risks

There are a number of risks that have been highlighted, many of which have dependencies. These must be carefully considered to ensure success for the project. Key risks are listed below in Table 7.

Table 7 Project Risks

Risk Area	Description
System Integration	
Glencore SAP Adaptors	RPM strongly recommends a direct integration with SAP which has previously been done at many sites across the globe. The transfer of reliable, accurate and timely data is critical to the success of the project and AMT. Given Glencore's current mandate to not allow direct SAP integration, Glencore will need to write its own adaptors.
Solution	
Operation	Latency introduced by SAP data stewards based in South Africa, and the ability of the data stewards to work with AMT data which may be in Spanish.
Multi-company	AMT was not designed to be used by two separate companies on a single instance; AMT has not previously been deployed to support two separate companies on a single instance. Whilst RPM has endeavoured to validate known aspects of the system which may prevent AMT operating as per Glencore's single instance request during the 2 day workshop, it is quite feasible that issues may be identified during the course of the deployment project which prevents a single instance solution. Glencore and RPM openly acknowledge this potential eventuallity. If this transpires, both parties agree that the solution will revert to two separate instances (one for LB and one for AP) and will not attempt in any way to change the core design of AMT. This may impact the project timeline, scope and budget.
Time Zones	AMT uses the server time to capture data and timestamp. AP and LB are on different time zones so one or both sites will need to compromise on the date and time used.
Configuration	The 2 day workshop focussed on the typical Functionality-level, System-level, Security-level, Integration-level and System Table-level configurations which impact AMT's ability to adequately support a single instance for two operations which are, in fact, two separate companies operating in different countries. Not all configurations were reviewed particularly with respect to integration. Integration will require dedicated integration design workshops (covering SAP, MM and PI). It is quite feasilbe that during the subsequent detailed solution and/or integration design workshops that a configuration requirement for LB or AP, which cannot be standardised, may necessitate two separate AMT instances. This may impact the project timeline, scope and budget.
IT Architecture	
Network Performance	Glencore acknowledges that AMT's performance and responsiveness may be compromised due to line-speed issues by Glencore's decision to: 1. Implement a single instance of AMT for LB and AP, and 2. Locate the AMT Production server in Sanitago. If poor line-speed issues cannot be resolved, this may necessitate implementing separate AMT servers on site at LB and AP respectively. This may impact the project timeline, scope and budget.
Business	

Risk Area	Description
Standardisation	Moving disparate operations to a global combined standard with respect to systems, data, business processes, organisational responsibilities, KPIs, etc. is a significant undertaking. Glencore is responsible to design, develop, enact, embed and foster acceptance of any and all standards. Glencore is responsible for data standardisation, conversion of existing data to the new standard, data quality and data cleansing. AMT's ability to operate successfully, particualarly on a single platform with a standard configuration for both sites, will depend on Glencore's ability to successfully implement the new standards and make the changes "stick" long term. Glencore implementing a robust and responsive stewardship process and master data group; empowered to ensure implementation agreements made to support a single instance are not eroded after go-live. Complete alignment of reporting requirements across both sites and departments within sites.
Change Management	The specific change management item (above) notwithstanding, Glencore is responsible for the change management of all aspects of AMT pre- and post-deployment across both operations as though they are a single company, e.g. helpdesk support, user training, deploying additional functionality, upgrades , licensing and maintenance, etc.
Implementation Project	
Timeline	Glencore want both LB and AP to Go-live on a single instance of AMT by 1st January 2020. At the time of writing this document, this is just over a 4 month duration for project completion. SAP is scheduled to Go-live in parallel at AP on 1st January 2020. A typical AMT deployment project, for a single operation with its own instance of AMT, is generally 6 months from the date the kick-off workshop commences. The LB implementation project plan developed in March 2019 was shown to the team in the workshop as an example. That timeline showed a duration requirement of greater than 4 months. This project coincides with the Christmas and New Year holiday period. RPM has advised Glencore that achieving a 1st January 2020 Go-live date is not achievable for either LB or AP. It is expected that AP will need to consider following the same approach taken by LB when Ellipse was turned off.
Resources	Defining and implementing new standards across two operations at the same time takes resources. Implementing AMT at two sites at the same time takes resources. Undertaking both of these activities at the same time at AP, which is also devoting resources to an SAP implementation, will be challenging and introduces significant risk. The SAP project may potentially absorb AMT project resources and users on short-notice for an extended period. At the conclusion of the August 2019 workshop, the Glencore team was still seeking senior management's approval to secure project resources. Glencore acknowledges that this is key to Glencore and the AMT project team possibly accomplishing the activities described in this document in a reasonable timeframe.

6. Approval Section

Table 8 comprises a list of the major stakeholders for the project and their agreement with and approval of this AMT Solution Design.

Table 8 AMT Solution Design Approval

Name	Role	Company	Date	Signature
		Lomas Bayas		
		Antapaccay		